

Toan Nguyen

Los Angeles, California | tientoan@usc.edu | toannguyen1904.github.io

Education

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- University of Southern California**, Ph.D. in Computer Science Aug 2025 – Present
- Advisor: **Prof. Yue Wang** and **Prof. Daniel Seita**
- Ho Chi Minh City University of Science**, BS in Computer Science Sep 2019 – March 2024
- GPA: 3.64/4.00, 8.49/10.00
 - Thesis: Open-Vocabulary Affordance Detection in 3D Point Clouds. Grade: **11/10**. Advisor: **Prof. Bac Le**

Research Experience

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- AI Research Resident**, FPT Software AI Center, Ho Chi Minh Jul 2022 – Apr 2025
- Robotics and Computer Vision Team. Advisors: **Prof. Anh Nguyen** and **Prof. Ngan Le**
 - Research topics: Multimodal Robot Learning, Generative Models for Robotics and 3D Scene Generation, Embodied Navigation, LLM-Based Multi-Agent Systems
- Research Assistant**, Ho Chi Minh City University of Science, Ho Chi Minh Jan 2022 – Jul 2022
- AISIA Research Lab. Advisor: **Prof. Binh Nguyen**
 - Research topics: Representation Learning, Fine-Grained Image Classification, Insect Pest Classification and Detection, Image Data Augmentation

Publications

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- Language-Driven 6-DoF Grasp Detection Using Negative Prompt Guidance**
Toan Nguyen, Minh Nhat Vu, Baoru Huang, An Vuong, Quan Vuong, Dung Nguyen, Ngan Le, Thieu Vo, Anh Nguyen
European Conference on Computer Vision (ECCV), 2024, **Oral Presentation (top 2.33%)**
- HabiCrowd: A High-Performance Simulator for Crowd-Aware Visual Navigation**
An Vuong, **Toan Nguyen**, Minh Nhat Vu, Baoru Huang, Dung Nguyen, Binh Huynh, Thieu Vo, Anh Nguyen
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024
- Language-Conditioned Affordance-Pose Detection in 3D Point Clouds**
Toan Nguyen, Minh Nhat Vu, Baoru Huang, Tuan Vo, Vy Truong, Ngan Le, Thieu Vo, Bac Le, Anh Nguyen
IEEE International Conference on Robotics and Automation (ICRA), 2024
- Open-Vocabulary Affordance Detection Using Knowledge Distillation and Text-Point Correlation**
Tuan Vo, Minh Nhat Vu, Baoru Huang, **Toan Nguyen**, Ngan Le, Thieu Vo, Anh Nguyen
IEEE International Conference on Robotics and Automation (ICRA), 2024
- Language-Driven Scene Synthesis Using Multi-Conditional Diffusion Model**
An Vuong, Minh Nhat Vu, **Toan Nguyen**, Baoru Huang, Dung Nguyen, Thieu Vo, Anh Nguyen
Conference on Neural Information Processing Systems (NeurIPS), 2023
- Open-Vocabulary Affordance Detection in 3D Point Clouds**
Toan Nguyen, Minh Nhat Vu, An Vuong, Dung Nguyen, Thieu Vo, Ngan Le, Anh Nguyen
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023, **Best Overall & Best Student Paper Awards Finalist (top 0.43%)**

Presentations

Oral paper presentation, ECCV, 2024

Milano, Italy

Oral paper presentation, IROS, 2023

Michigan, USA

Honors and Awards

- **Viterbi Fellowship**, University of Southern California, 2025
- **Registration Grant**, ECCV, 2024
- **Oral Presentation**, ECCV, 2024
- **Award for Technology and Innovation Publications**, Ho Chi Minh City University of Science, 2024
- **Best Paper & Best Student Paper Awards Finalist**, IROS, 2023

Professional Services

- **Conference Reviewer:** WACV 2026, IROS 2025, ECCV 2024, ICRA 2024, IROS 2024
- **Journal Reviewer:** RA-L 2025

Skills

Programming: Python, C/C++, Julia, R, Java, Javascript, SQL, HTML, CSS, Linux, MATLAB

AI: Robotics, Generative Models, Embodied AI, 2&3D Computer Vision, Multimodal Learning, Large Language Models, Multi-Agent Systems

Frameworks: Pytorch, Tensorflow, Keras, Jax, Pandas, NumPy, scikit-learn, OpenCV, ROS, Node.js, React

Tools: Docker, Google Cloud, Jupyter Notebook/Lab, Git, Blender

Languages: English, Vietnamese